

AgroShield AI

Defense report: feasibility of procedure, architecture, equipment, data collection and service delivery

Scope: farm-level salinity-risk intelligence for smallholder farms in coastal Bangladesh, demonstrated first on one district and one crop (for example boro rice). All quantities, frequencies and costs below are planning assumptions to be confirmed with local suppliers, partners and pilot data.

1. Feasibility of the procedure

The procedure follows the loop **Farm data, prediction, explanation, decision support, farmer action**. Each step is feasible because it depends on tools and partners that already exist, and each step has a check before the next begins.

Step	What happens	Why it is feasible	Evidence gate
1. Register farm	Officer records location, boundary, crop, planting date, water source	Smartphone with GPS; extension officers already visit farms	100% of pilot farms have a valid boundary and crop record
2. Collect data	Field EC and water readings plus satellite and weather feeds	Handheld meters are basic tools; Sentinel and rainfall data are free	Data-sufficiency score per farm above threshold
3. Predict risk	Model outputs risk level, confidence and data sufficiency	Tabular ML on modest data; baselines first, then candidates	Beats a simple threshold baseline on held-out data
4. Explain	Top contributing factors shown as evidence, not causation	Standard explainability methods for tabular models	Experts confirm explanations are sensible
5. Recommend	Expert-validated rules matched to crop and growth stage	Rules are written by agronomists, not generated by AI	Every rule signed off and versioned
6. Communicate	Bengali message by SMS, voice or officer visit	Existing mobile channels; generative AI only rephrases approved text	Farmer comprehension test in pilot
7. Correct	Compare with later readings; farmer flag; monitor drift	Same field visits supply ground truth	False-alarm and missed-warning rates tracked

Defense point: the procedure does not depend on any unproven technology. Its main risk is data volume, which the pilot is designed to build, and the staged funding releases money only after the gates above are met.

2. System architecture

The system is layered so each part can be tested, replaced or corrected independently. Data flows upward; feedback flows back down to retraining and rule review.

Layer	Components	Role
7. Delivery	Farmer SMS / voice / app, officer app, expert console, organization dashboard	How each user receives or reviews output
6. Communication	Generative AI (Bengali and English) constrained to approved recommendation text	Simplifies language; cannot add advice
5. Decision support	Rule engine: crop, growth stage, risk level, farm context; expert-versioned	Produces controlled recommendations
4. Climate Risk Engine	ML salinity model, confidence, data-sufficiency check, explainability	Predicts and explains farm-level risk
3. Farm Context Model	Per-farm store: boundary, soil, water, weather, crop, satellite, history	Makes data farm-specific

Layer	Components	Role
2. Ingestion and quality	Pipelines, unit checks, outlier flags, gap handling, geospatial alignment	Cleans and joins sources
1. Data sources	Field meters, sentinel sensors, weather stations, Sentinel-2 / Landsat, rainfall grids, partner records	Raw evidence
Cross-cutting	Monitoring (drift, accuracy), model and rule versioning, audit log, consent and access control	Correction and governance

Design choices that support feasibility:

- Cloud hosting with a small managed stack, so no local data centre is needed.
- SMS and voice fallback so farmers without smartphones or steady data are still reached.
- Human-in-the-loop: experts can suspend any rule or override an alert; high-severity cases go to an officer.
- Modular models: the salinity engine can be retrained or swapped without changing the farmer interface.

3. Equipment and its arrangement

Equipment is arranged in three tiers so the pilot gets rich data from a few farms and useful coverage from all farms, at a cost smallholder programs can afford.

Tier	Where	Equipment	Purpose
A. Sentinel farms (about 10 of 100)	Representative soil, water source and elevation classes	Soil EC / moisture / temperature probes at 2 depths with data logger and cellular or LoRa link; water-source EC meter	Continuous ground truth to train and calibrate the model
B. Cluster level (about 10 clusters of 10 farms)	One point per cluster, central to farms	Rain gauge or small weather station; a field officer with handheld EC meter, GPS phone or tablet, sample kit, calibration solutions	Local weather and scheduled EC readings for every farm
C. All farms	Remote	Sentinel-2 / Landsat indices, gridded rainfall and temperature, tide or river-salinity data from partners	Low-cost coverage and context
Back end	Cloud	Managed database, compute for training, storage, SMS / voice gateway, dashboards	Run the platform and deliver services

Arrangement rules:

- Place sentinel farms across the range of soil type, distance to saline water, and elevation, not only in easy-to-reach fields.
- Measure at a fixed georeferenced point and fixed depth on every visit, so readings are comparable over time.
- Calibrate meters against standard solution at the start of each field round and log the calibration.
- Site cluster weather stations away from trees and buildings; cross-check against gridded rainfall.
- Keep spare meters and a repair route with the partner NGO to avoid data gaps.

4. Data collection

Data	Source	Frequency (planning)	Quality control
Soil EC, moisture	Handheld meter (all farms); sentinel probes (10 farms)	Every 2 weeks in season, monthly off-season; probes hourly	Calibration log, duplicate reading, range checks
Water salinity	Handheld meter at irrigation / canal / pond	Every 2 weeks in season; after tide or flood events	Same protocol; record source and time
Weather	Cluster station and gridded products	Daily	Compare station with grid; flag gaps

Data	Source	Frequency (planning)	Quality control
Satellite indices	Sentinel-2, Landsat	Every 5 to 10 days when cloud-free	Cloud masking; validate against field EC before use
Crop and management	Officer registration and farmer report	At planting, stage changes, harvest	Standard crop and stage lists
History and outcomes	Partner records, farmer recall, yield notes	Each season	Cross-check with officer visit

Defining the target: risk levels are set with agronomists using crop-specific EC tolerance thresholds and observed yield or stress outcomes, then tested for consistency. The prediction horizon (for example 1 to 4 weeks ahead) is fixed before modelling.

Data sufficiency: each prediction carries a score for how recent and complete its inputs are. If it is low, the system shows a wider range or asks for a new reading instead of a confident label.

Governance: farmers give informed consent; personal data is separated from analysis data; only aggregated results go to organizations; data ownership and sharing terms are agreed with partners in writing.

5. Service provision

User	What they receive	Channel	Who supports
Farmer	Plain-language risk level, main reasons, confidence, suggested actions for their crop and stage	Bengali SMS, voice call, officer visit, optional app	Extension officer or NGO field agent
Field officer	Farm list, visit schedule, data entry, alerts needing follow-up	Mobile app (works offline, syncs later)	Cluster supervisor
Agronomist / expert	Rule library, flagged predictions, override and suspend controls	Web console	Project agronomy lead
Organization / government	District risk map, trends, priority areas, pilot metrics	Dashboard and periodic report	Project data team

Service flow: new reading or data update, risk update, alert rule check, expert-approved message, delivery to the farmer, follow-up reading, outcome logged. Urgent alerts are also sent to the responsible officer, who can call or visit.

Service targets (proposed for the pilot): alerts sent within 24 hours of a risk change; every high-risk alert followed up by an officer within 3 days; monthly review of false alarms and missed warnings; each rule reviewed by an agronomist before release.

Sustainability: farmers receive the service free or at very low cost; NGOs, government programs, insurers and agribusiness fund it in exchange for aggregated risk intelligence, with grants or donor support covering the gap early on.

6. Risks and limits

Risk	Mitigation
Too little farm-level EC data to train a reliable model	Sentinel probes plus scheduled readings; start with baselines; report accuracy with uncertainty; expand only after validation
Satellite pixels blur small plots or are blocked by monsoon cloud	Use satellite as context, not sole input; add cluster weather and field EC
Meter error or missed visits	Calibration logs, duplicate readings, spare equipment, data-sufficiency warnings
Farmers ignore or misread alerts	Officer-mediated delivery, simple Bengali wording, comprehension testing

Risk	Mitigation
Model degrades in a new season or region	Drift monitoring, shadow testing, retraining triggers, fresh validation for each new area
Bad advice reaches farmers	Expert-validated rules only; versioned, suspendable, traceable

7. Anticipated judge questions

Question	Short answer
Why is this feasible with limited data?	It starts small, builds its own dataset in the pilot, and uses hybrid sensing: a few continuous sensors, scheduled field readings for all farms and free satellite data for coverage.
Why not just use fixed thresholds?	Thresholds are the baseline. The model is kept only if it beats them on held-out data, and it adds confidence and explanations.
Who pays for equipment?	Handheld meters and a few probes are the main capital cost. Partners and grants fund the pilot; institutional buyers fund scale-up.
How do you avoid harmful advice?	Generative AI cannot create advice. Only expert-validated, versioned rules are sent.
What if farmers lack smartphones?	SMS, voice and officer visits carry the service; the app is optional.
How will you prove it works?	A 100-farm pilot measuring accuracy, warning lead time, false-alarm rate, comprehension, adoption and outcomes.

Note: figures such as sensor counts, visit frequencies and service targets are proposals for the pilot design, not measured results. They should be refined with local partners and equipment quotes before submission.